SEO Keyword Research & Content Optimization System

Project Report | Digital Marketing Internship (Day 40-43)

Intern: Sadhanala Nischay Prasad

Domain: Digital Marketing

Target website: NexaByte Solutions - a sample IT services website (web, app, AI and UI/UX services) based in Hyderabad

Tools: Google Search, Google Trends, Google Keyword Planner concepts, Google Search Console concepts, Google Sheets / Excel, HTML, Google Docs

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1. Introduction & Objective

Search engine optimization (SEO) helps a website appear in Google when potential customers search for its services. This project builds a complete, repeatable SEO system for a technology services website: find the right keywords, understand why people search them, map each keyword to the right page, fix on-page issues and plan content that brings steady organic traffic.

Objective: learn practical SEO fundamentals - keyword research, search intent, on-page optimization, content structure, internal linking and SEO measurement.

2. Methodology

• **Seed keywords:** listed the site's core services (web development, app development, AI solutions, UI/UX) plus location (Hyderabad).

• **Expansion:** Google autocomplete, 'People also ask' and 'Related searches' to find long-tail variations.

• **Volume:** estimated using Google Keyword Planner volume ranges (10-100, 100-1K, 1K-10K, India).

• **Trends:** Google Trends (India, last 12 months) to mark each keyword Rising or Stable.

• **Difficulty:** manual 0-100 score from page-1 review - presence of big brands and directories, number of ads and SERP features.

• **Prioritization:** Priority Score = log10(Volume) x 10 x (100 - KD)/100 x Intent weight (Transactional 1.0, Commercial 0.9, Informational 0.6, Navigational 0.4).

• **Audit:** page-by-page review of titles, meta descriptions, headings, content, images, URLs, links and schema.

Note: volumes and difficulty are estimates for learning purposes and should be verified with live tool data before real use.

3. Keyword Research Dataset

The dataset contains **33 keywords** across 7 clusters with a combined estimated volume of **25,100 searches/month** and an average difficulty of **38.3**. The full dataset is in *SEO_Keyword_Research_System.xlsx* and *data/keyword_research_dataset.csv*.

Top 12 keywords by Priority Score

Keyword	Intent	Volume	KD	Target page	Score
website development cost in india	Commercial	1300	35	/blog	18.2
ecommerce website development hyderabad	Transactional	390	40	/web-development	15.5
ui ux design company hyderabad	Commercial	480	38	/ui-ux-design	15.0
ai chatbot development services	Commercial	590	41	/ai-solutions	14.7
website redesign services	Commercial	320	35	/ui-ux-design	14.7
custom ai solutions for business	Commercial	260	33	/ai-solutions	14.6
digital solutions agency hyderabad	Commercial	140	25	/	14.5
flutter app development company	Commercial	590	42	/mobile-app-development	14.5
hire app developers india	Transactional	480	47	/mobile-app-development	14.2
software company near me	Transactional	1900	57	/contact	14.1
ai development company in india	Commercial	880	48	/ai-solutions	13.8
generative ai consulting services	Commercial	320	39	/ai-solutions	13.8

4. Primary & Secondary Keywords

Page	Primary keyword	Secondary / long-tail
Home	software development company in hyderabad, nexabyte solutions	it services company hyderabad, digital solutions agency hyderabad
Web Development	web development company in hyderabad	website development services, website design company hyderabad, custom website development, best web development company in hyderabad, ecommerce website development hyderabad
Mobile App Development	mobile app development company in hyderabad	android app development services, flutter app development company, hire app developers india
AI Solutions	ai development company in india	ai chatbot development services, generative ai consulting services, custom ai solutions for business
UI/UX Design	ui ux design company hyderabad	ui ux design services, website redesign services
Contact	-	nexabyte solutions contact, software company near me

5. Search-Intent Classification

Intent	Count	Meaning / content	Examples
Transactional	3	Ready to act - strong CTA, contact/quote	ecommerce website development hyderabad, hire app developers india, software company near me
Commercial	19	Comparing providers - service page with proof	software development company in hyderabad, it services company hyderabad, digital solutions agency hyderabad, web development company in hyderabad ...
Informational	9	Wants to learn - blog/guide	how much does a website cost in india, app development cost in india, how to build a mobile app for my business, ai automation for small business ...
Navigational	2	Looking for our brand - home/contact	nexabyte solutions, nexabyte solutions contact

Key insight: commercial keywords belong on service pages, while informational keywords (cost guides, comparisons) need blog content that links back to the service pages.

6. Keyword Difficulty Analysis

Band	Count	Strategy
Easy (KD < 30)	10	Quick wins - publish blog posts first
Medium (30-49)	13	Target with strong service pages + internal links
Hard (50+)	10	Long-term - build authority, reviews, backlinks, local SEO

7. Competitor Keyword Analysis

Top Google results for key terms were reviewed (incognito, India) to see what type of pages rank and where the site has gaps.

Keyword	Who ranks	SERP features	Our gap
web development company in hyderabad	Local agency service pages + directories (Clutch/GoodFirms lists)	Local pack, People Also Ask	No city keyword in our title/H1; add portfolio + FAQ + LocalBusiness schema
software development company in hyderabad	Directory listicles + large IT firm homepages	Local pack, Ads	Home page H1 is generic; add Hyderabad, trust signals, links to each service
mobile app development company in hyderabad	Agency service pages	PAQ, Ads	No case studies or tech-stack section on our page
ai chatbot development services	National agencies + SaaS vendors	PAQ, Videos	Our AI page has no use-case breakdown; add industry use cases
ui ux design company hyderabad	Design studio pages	Images, Local pack	Thin page (~200 words); add process + portfolio
website development cost in india	Blog guides + pricing pages	Featured snippet (table), PAQ	No content; win snippet with a clear cost table
app development cost in india	Blog guides from agencies	Featured snippet, PAQ	No content; create guide + simple cost estimator
static vs dynamic website	Educational articles	Featured snippet (table)	Easy topical win; comparison table + link to /web-development
react vs next js which is better	Dev blogs, docs, YouTube	Videos, PAQ	Write practical comparison from agency viewpoint
software company near me	Google Maps / local pack heavy	Local pack dominant	Needs Google Business Profile + NAP on /contact

8. Keyword-to-Page Mapping

Each page targets one primary keyword to avoid keyword cannibalization.

URL	# kw	Keywords
/	4	software development company in hyderabad, it services company hyderabad, digital solutions agency hyderabad ...
/web-development	6	web development company in hyderabad, website development services, website design company hyderabad ...
/mobile-app-development	4	mobile app development company in hyderabad, android app development services, flutter app development company ...
/ai-solutions	4	ai development company in india, ai chatbot development services, generative ai consulting services ...
/ui-ux-design	3	ui ux design company hyderabad, ui ux design services, website redesign services
/contact	2	nexabyte solutions contact, software company near me
/blog	10	website development cost in india, how much does a website cost in india, app development cost in india ...

9. On-Page SEO Audit

Page	Element	Issue	Recommendation	Priority
/	Title tag	'Home | NexaByte' - no keyword, wasted space	Use primary keyword + city + brand (50-60 chars)	High
/	H1	'Welcome to Our Website' - generic	H1 with primary keyword: Software Development Company in Hyderabad	High
/	Meta description	Missing - Google auto-generates snippet	Write 140-160 char description with value prop + CTA	High
/	Internal links	Service cards are not linked	Link each service card to its service page with descriptive anchor	Medium
/web-development	Title tag	'Services' - duplicate across 4 pages	Unique title per page targeting its primary keyword	High
/web-development	Heading structure	Two H1s; H4 used before H2	One H1, logical H2 > H3 order	High
/web-development	Content depth	~250 words, no FAQ or process	Expand to 900-1,200 words: services, process, tech stack, FAQ	High
/web-development	Images	Hero image 2.4 MB, alt text missing	Compress to WebP (<150 KB), descriptive alt text	Medium
/web-development	URL	/services?id=2	Clean URL /web-development + 301 redirect	Medium
/mobile-app-development	Content	No case studies, no tech stack	Add Flutter/React Native section + 2 case studies	Medium
/ai-solutions	Title/H1	Keyword 'AI' only, no service term	Target 'AI development company in India'	High
/ui-ux-design	Content	Thin content (~200 words)	Add design process, tools, portfolio	Medium
/about	E-E-A-T	No team, no founding year, no address	Add team, experience, office address	Low
/contact	Local SEO	Address not in text, no map, no schema	Add NAP, Google Map embed, LocalBusiness schema	High
Sitewide	Schema markup	None	Organization + LocalBusiness + FAQPage schema	Medium
Sitewide	Canonical tags	Missing	Self-referencing canonical on every page	Medium
Sitewide	Sitemap / robots	No XML sitemap submitted	Create sitemap.xml, submit in Search Console	High
Sitewide	Mobile / speed	Render-blocking CSS/JS	Defer non-critical JS, lazy-load images	Medium

10. Title & Meta-Description Recommendations

Titles kept within 30-60 characters and meta descriptions within 120-160 characters (lengths checked with LEN() in the sheet).

Page	Recommended title	Recommended meta description
/	Software Development Company in Hyderabad | NexaByte (52)	NexaByte builds websites, mobile apps and AI solutions for growing businesses in Hyderabad. Get a free project estimate today. (126)
/web-development	Web Development Company in Hyderabad | NexaByte (47)	Custom website development services in Hyderabad - fast, SEO-friendly and mobile-ready sites for startups and SMEs. Request a free quote. (137)
/mobile-app-development	Mobile App Development Company in Hyderabad | NexaByte (54)	Android, iOS and Flutter app development in Hyderabad. From idea to Play Store launch, with transparent pricing. Talk to our team. (130)
/ai-solutions	AI Development Company in India | Chatbots & Automation (55)	Custom AI solutions, chatbot development and generative AI consulting to automate your business. See use cases and book a demo. (127)
/ui-ux-design	UI UX Design Company in Hyderabad | NexaByte (44)	User-friendly UI/UX design and website redesign services. Research, wireframes and prototypes that convert visitors into customers. (131)
/about	About NexaByte | Software Team in Hyderabad (43)	Meet the NexaByte team - developers and designers in Hyderabad building web, app and AI products for businesses across India. (125)
/contact	Contact NexaByte | Software Company in Hyderabad (48)	Call, email or visit NexaByte in Hyderabad. Share your project idea with our team and get a free, no-obligation estimate within 24 hours. (137)

11. Heading Structure Recommendations

Rule: exactly one H1 per page containing the primary keyword, H2 for main sections, H3 for sub-points - never skip levels.

Page	Tag	Heading
/	H1	Software Development Company in Hyderabad
/	H2	Web, App & AI Services for Growing Businesses
/	H3	Web Development / Mobile App Development / AI Solutions / UI/UX Design
/	H2	Why Businesses in Hyderabad Choose NexaByte
/	H2	Recent Projects
/	H2	Frequently Asked Questions
/web-development	H1	Web Development Company in Hyderabad
/web-development	H2	Our Website Development Services
/web-development	H3	Custom Website Development / E-commerce Websites / Website Redesign
/web-development	H2	Our Development Process
/web-development	H2	Technologies We Use
/web-development	H2	How Much Does a Website Cost?
/web-development	H2	FAQs About Website Development
/mobile-app-development	H1	Mobile App Development Company in Hyderabad
/mobile-app-development	H2	Android, iOS & Flutter App Development
/mobile-app-development	H2	Case Studies
/mobile-app-development	H2	Hire App Developers
/ai-solutions	H1	AI Development Company in India
/ai-solutions	H2	AI Chatbot Development Services

Page	Tag	Heading
/ai-solutions	H2	Generative AI Consulting
/ai-solutions	H2	AI Use Cases by Industry
/ui-ux-design	H1	UI UX Design Company in Hyderabad
/ui-ux-design	H2	Our Design Process
/ui-ux-design	H2	Website Redesign Services

12. Internal-Linking Plan

Hub-and-spoke model: service pages are hubs; blog posts are spokes linking back with descriptive anchor text (no 'click here').

From	To	Anchor text
/	/web-development	web development company in Hyderabad
/	/mobile-app-development	mobile app development
/	/ai-solutions	AI development services
/	/ui-ux-design	UI/UX design services
/web-development	/ui-ux-design	website redesign services
/web-development	/blog/website-development-cost-india	website development cost in India
/web-development	/contact	get a free website quote
/mobile-app-development	/blog/app-development-cost-india	app development cost in India
/mobile-app-development	/ai-solutions	AI-powered app features
/ai-solutions	/blog/ai-automation-small-business	AI automation for small businesses
/ui-ux-design	/web-development	custom website development
/blog/website-development-cost-india	/web-development	website development services
/blog/app-development-cost-india	/mobile-app-development	mobile app development company in Hyderabad
/blog/static-vs-dynamic-website	/web-development	custom website development
/blog/react-vs-nextjs	/web-development	web development services
/blog/ai-automation-small-business	/ai-solutions	custom AI solutions
/blog/benefits-of-website-small-business	/blog/website-development-cost-india	how much a website costs
/about	/contact	contact our team

13. SEO Content Ideas (6-week plan)

Week	Article	Target keyword	Links to
Week 1	Website Development Cost in India (2026 Price Guide)	website development cost in india	/web-development
Week 1	App Development Cost in India: Breakdown by App Type	app development cost in india	/mobile-app-development
Week 2	Static vs Dynamic Website: Which One Does Your Business Need?	static vs dynamic website	/web-development
Week 2	React vs Next.js: Which Is Better for Your Project?	react vs next js which is better	/web-development
Week 3	10 Benefits of Having a Website for Small Businesses	benefits of website for small business	/web-development
Week 3	How Long Does It Take to Build a Website? (Timeline)	how long does it take to build a website	/web-development
Week 4	AI Automation for Small Business: 8 Practical Use Cases	ai automation for small business	/ai-solutions
Week 4	How to Build a Mobile App for Your Business: Step-by-Step	how to build a mobile app for my business	/mobile-app-development
Week 5	What Is an SEO-Friendly Website? Checklist for 2026	what is an seo friendly website	/web-development
Week 5	How to Choose the Best Web Development Company in Hyderabad	best web development company in hyderabad	/web-development
Week 6	Flutter vs Native App Development: Cost & Performance	flutter app development company	/mobile-app-development

Week	Article	Target keyword	Links to
Week 6	AI Chatbots for Business: Costs, Benefits & Examples	ai chatbot development services	/ai-solutions

14. Implementation Example (HTML)

The Web Development page was rewritten in HTML (see *html/optimized-web-development.html*). Changes: keyword-focused title and meta description, one H1, logical H2/H3, descriptive alt text with WebP image and lazy loading, canonical tag, Open Graph tags, LocalBusiness schema, breadcrumb, FAQ section and contextual internal links.

Element	Before	After
Title	Services	Web Development Company in Hyderabad | NexaByte
H1	Two H1s ('Welcome', 'Our Services')	One H1 with primary keyword
Meta description	Missing	137-char description with CTA
Image	No alt, 2.4 MB PNG	Descriptive alt, WebP, lazy-loaded
Links	'Click here', /services?id=5	Descriptive anchors, clean URLs
Schema	None	ProfessionalService JSON-LD

15. SEO Measurement Plan

KPI	Where to measure	3-month target
Impressions	Search Console > Performance	+50% in 3 months
Clicks	Search Console > Performance	+40% in 3 months
CTR	Search Console > Performance	> 3% sitewide
Average position	Search Console > Performance	Primary keywords in Top 10
Indexed pages	Search Console > Pages	100% of important pages
Core Web Vitals	Search Console > Core Web Vitals	All URLs 'Good'
Organic leads	Contact form / Analytics	10+ enquiries per month
Keyword rankings	Manual Google Search check (incognito)	Weekly log in Sheets

Process: submit sitemap in Search Console, record a baseline in week 0, then log impressions, clicks, CTR and average position weekly in Google Sheets. Pages with high impressions but low CTR get title/meta rewrites first.

16. Key Learnings

• Search intent matters more than volume - a 400-search commercial keyword can bring more leads than a 2,000-search informational one.

• One page should target one primary keyword; mapping prevents pages competing with each other.

• Low-difficulty long-tail keywords are the fastest wins for a new or small website.

• Titles, meta descriptions and headings are small edits with large impact on CTR and relevance.

• Internal linking connects blog traffic to money pages and helps Google understand site structure.

• SEO is measurable: Search Console data shows what to improve next.

17. Conclusion

This project produced a complete SEO system: a prioritized keyword dataset, intent and difficulty analysis, competitor gaps, keyword-to-page mapping, an on-page audit with fixes, title/meta and heading recommendations, an internal-linking plan, a 6-week content plan and a measurement framework. Applied together, these steps give a technology services website a clear roadmap to rank for local commercial searches and grow organic leads.